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ACX7100/ACX7509 Validation: L3VPN/6VPE

An HPE Juniper Networking Techpost

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1 L3VPN Validation on ACX7000

Suneesh Babu - 11/09/2022

ACX7000 platform has been tested successfully with 4,000 Layer3 VPN Routing-instances with BGPv4, BGPv6, OSPF, OSPFv3, ISISv4, ISISv6, Static-v4, Static-v6 as CE-PE protocols and with a total of 1.35M routes.

1.1 Introduction

This is the third article in the ACX7k Metro Validation Series.

In this article we validate the L3VPN, 6VPE, 6PE feature scale support on ACX7100-32C platform with 22.2R1 Junos-EVO build.

RFC 4364 describes BGP-enabled L3VPN to connect Customer Edge Devices (CEs) across a Provider Edge (PE) Network. The PEs provide layer 3 connectivity for their customers using various protocols like BGP, OSPF, ISIS, RIP or Static. The PEs of the network are connected via MPLS technology LDP, RSVP, SR-MPLS etc. In L3VPN, the IP address associated with the hosts are learned via the control plane on both CE-PE side and across PE network with the help of BGP.

JUNOS-EVO supports L3VPN and 6VPE functionality with instance-type “vrf”. In this document, we qualify the feature scale on the ACX7000 platforms.

ACX7100-32C with 22.2R1 Junos Version supports different flavours of CE-PE. The Key Performance Indicators (KPI) tested are captured in the below table. The same scale is supported and tested on ACX7100-48L as well as on ACX7509. But the ACX7024 Scale is not covered in this article (same features but different scale are expected on this platform).

Please note that despite sharing the same ACX moniker, the ACX7000 products are different products than ACX500/710/1000/1100/2100/2200/4000/5000/5400/6000. They are powered by different Packet Forwarding Engines (PFE), and support different feature sets and scales.

KPI	Scale
L3VPN Instances	4,000
6VPE Instances (Dual Stack)	1,150
BGPv4 Peering (CE-PE)	2,000
BGPv6 Peering (CE-PE)	500
OSPF Neighbors (CE-PE)	1,000
OSPFv3 Neighbors (CE-PE)	100
ISIS IPv4 Adjacencies (CE-PE)	500
ISIS IPv6 Adjacencies (CE-PE)	500
Static-v4 Peering (CE-PE)	500
Static-v6 Peering (CE-PE)	50
VPNv4 Routes	1,151,000

KPI	Scale
VPNv6 Routes	115,000
IPv4 Global Routes	60,000
IPv6 Global Routes (6PE)	30,000
Total Route in the DUT	1,350,000

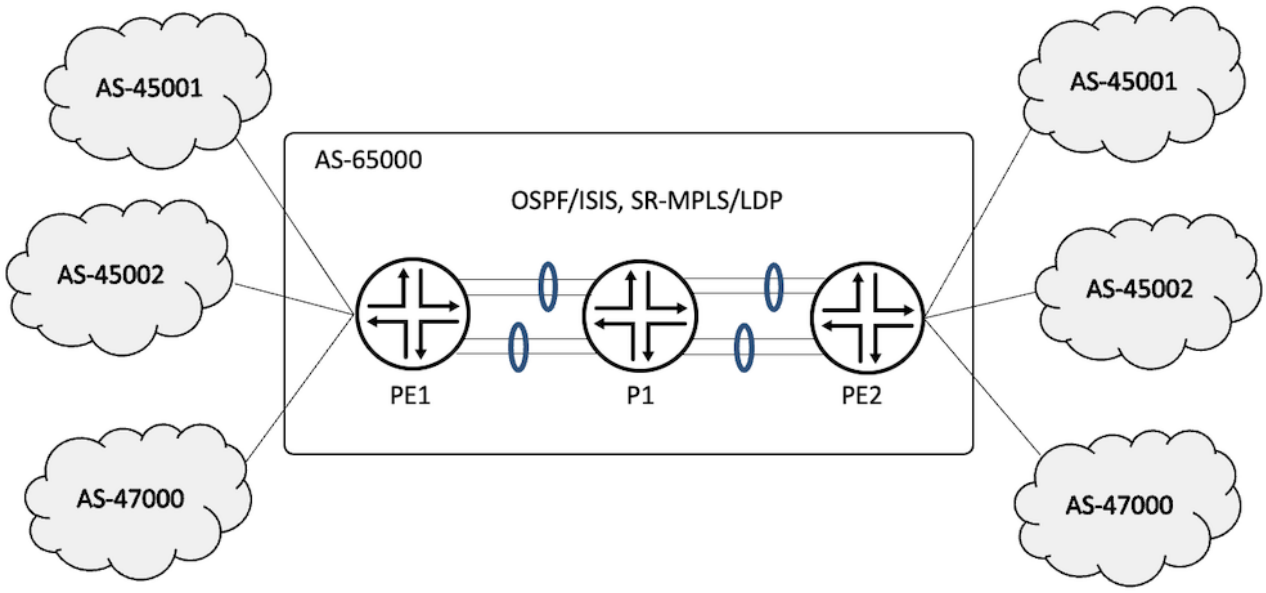
1.2 Metro L3VPN Topology

Test topology consists of three routers:

- an ACX7100-32C is the PE1
- an ACX7100-48L is the PE2
- a PTX10008 is the P1.

CEs are simulated using a traffic generator. PEs are connected to CEs using 5x 100G links and to the core using 8x 100G links, which are bundled in 2x LAGs each having four links.

The ACX7100-32C is the Device Under Test (DUT). The underlay transport used for testing is SR-MPLS with both OSPF and ISIS protocols individually. We also covered LDP as the underlay transport. The CE-bound interface configurations are of SP-Style (see glossary).



For this test, we configure 4,000 L3VPN instances:

- 2,000 have EBGPv4 as the CE-PE protocol,
- 1,000 have OSPF as the CE-PE protocol,
- 500 have ISIS as the CE-PE protocol
- the remaining 500 have Static-v4 reachability.
- 1,150 routing instances are
  - dual-stacked,

  - with 500 EBGPv6 Peering,
  - 100 OSPFv3 Peering,
  - 500 IS-IS Peering
  - 50 Static IPv6 reachability.

- The DUT learns 1.35M routes including VPNv4, VPNv6, global IPv4 and IPv6.

Bidirectional traffic in iMIX mode at 99.9% offer-load flows for all the L3VPN, 6VPE services. A total of ~1Tbps traffic transits through the DUT.

1.2.1 Base Configuration

The IGP and BGP configuration required to bring up L3VPN, 6VPE, 6PE are as follows:

```
regress@PE1> show configuration protocols |display inheritance no-comments
bgp {
   group IBGP {
      type internal;
      local-address 12.1.1.1;
      family inet {
         unicast;
      }
      family inet-vpn {
         unicast;
      }
      family inet6 {
         labeled-unicast;
      }
      family inet6-vpn {
         unicast;
      }
      neighbor 12.1.1.3 {
         export nhself;
      }
   }
   group EBGP {
      type external;
      as-override;
      neighbor 16.100.1.2 {
         family inet {
            unicast;
         }
         peer-as 65001;
      }
      neighbor 2002:16:100:1::2 {
         family inet6 {
            unicast;
         }
         peer-as 65001;
      }
   }
}
mpls {
   icmp-tunneling;
   ipv6-tunneling;
   interface ae0.0;
   interface ae1.0;
}
ospf {
   source-packet-routing {
      node-segment ipv4-index 101;
      srgb start-label 45000 index-range 2000;
   }
   area 0.0.0.0 {
      interface lo0.0 {
         passive;
      }
      interface ae0.0 {
         interface-type p2p;
      }
      interface ae1.0 {
         interface-type p2p;
```

```
        }
    }
}

regress@PE1>
```

The following configurations brings-up BGPv4 and BGPv6 Peering on a routing-instance.

```
interfaces {
    et-0/0/0 {
        description To_VPNv4_VPNv6_CE;
        flexible-vlan-tagging;
        mtu 9022;
        encapsulation flexible-ethernet-services;
        unit 1 {
            vlan-id 1;
            family inet {
                address 16.0.1.1/24;
            }
            family inet6 {
                address 2002:16:0:1::1/64;
            }
        }
    }
}
routing-instances {
    METRO_VRF_1 {
        instance-type vrf;
        protocols {
            bgp {
                group METRO_VRF_EBGPv4_1 {
                    type external;
                    family inet {
                        unicast;
                    }
                    peer-as 45001;
                    as-override;
                    neighbor 16.0.1.2;
                }
                group METRO_VRF_EBGPv6_1 {
                    type external;
                    family inet6 {
                        unicast;
                    }
                    peer-as 45001;
                    as-override;
                    neighbor 2002:16:0:1::2;
                }
            }
        }
        interface et-0/0/0.1;
        route-distinguisher 12.1.1.1:1;
        vrf-target target:65000:1;
        vrf-table-label;
    }
}
```

The following configuration brings up OSPF and OSPFv3 for a routing-instance

```
interfaces {
    et-0/0/2 {
        description To_VPNv4_OSPF_CE;
        flexible-vlan-tagging;
        mtu 9022;
        encapsulation flexible-ethernet-services;
        unit 1 {
            vlan-id 1;
            family inet {
                address 16.10.1.1/24;
            }
            family inet6 {
                address 2002:16:10:1::1/64;
            }
        }
    }
}
routing-instances {
    METRO_VRF_2001 {
        instance-type vrf;
```

```
        protocols {
           ospf {
              area 0.0.0.0 {
                 interface et-0/0/2.1 {
                    interface-type p2p;
                 }
              }
           }
           ospf3 {
              area 0.0.0.0 {
                 interface et-0/0/2.1 {
                    interface-type p2p;
                 }
              }
           }
        }
        interface et-0/0/2.1;
        route-distinguisher 12.1.1.1:2001;
        vrf-target target:65000:2001;
        vrf-table-label;
    }
}
```

Configuration to bring up ISISv4 and ISISv6 on a routing-instance:

```
interfaces {
   et-0/0/3 {
      description To_VPNv4_ISIS_CE;
      flexible-vlan-tagging;
      mtu 9022;
      encapsulation flexible-ethernet-services;
      unit 1 {
         vlan-id 1;
         family inet {
            address 16.14.1.1/24;
         }
         family iso;
         family inet6 {
            address 2002:16:14:1::1/64;
         }
      }
   }
   lo0 {
      unit 1 {
         family inet {
            address 12.2.0.1/32;
         }
         family iso {
            address 49.0001.0012.0002.0000.0001.00;
         }
         family inet6 {
            address 2005:12:2::1/128;
         }
      }
   }
}
routing-instances {
   METRO_VRF_3001 {
      instance-type vrf;
      protocols {
         isis {
            interface et-0/0/3.1 {
               level 1 disable;
               point-to-point;
            }
            interface lo0.1 {
               passive;
            }
         }
      }
      interface et-0/0/3.1;
      interface lo0.1;
      route-distinguisher 12.1.1.1:3001;
      vrf-target target:65000:3001;
      vrf-table-label;
   }
}
```

Configuration to bring up Staticv4 and Staticv6 for a routing-instance:

```
interfaces {
   et-0/0/4 {
      description To_VPNv4_Static_CE;
      flexible-vlan-tagging;
      mtu 9022;
      encapsulation flexible-ethernet-services;
      unit 1 {
         vlan-id 1;
         family inet {
            address 16.16.1.1/24;
         }
         family inet6 {
            address 2002:16:16:1::1/64;
         }
      }
   }
}
routing-instances {
   METRO_VRF_3501 {
      instance-type vrf;
      routing-options {
         rib METRO_VRF_3501.inet6.0 {
            static {
               route 2008:1:0:1::/64 next-hop 2002:16:16:1::2;
            }
         }
         static {
            route 46.0.1.0/24 next-hop 16.16.1.2;
         }
      }
      interface et-0/0/4.1;
      route-distinguisher 12.1.1.1:3501;
      vrf-target target:65000:3501;
      vrf-table-label;
   }
}
```

1.2.2 VPN CE Bound Interface Schema

The five CE bound 100G interfaces are logically split across the routing instances and detailed in this chart:

Metro L3VPN, 6VPE, 6PE and Global IPv4
Port
P-1
EBGPv6
P-2
Global v4
Global v6
P-3
OSPFv3
P-4
ISISv6
P-5
Staticv6

1.3 L3VPN Verification

The first step to verify the L3VPN service is to make sure that the IGP is established and routers in the core network are reachable. BGP Peering has to be verified for various services like L3VPN, 6VPE, 6PE and Global V4 routes. The L3VPN routes are present in bgp.l3vpn.0 table, 6VPE routes in bgp.l3vpn-inet6.0 table, 6PE routes in inet6.0 table and Global IPv4 routes in inet.0 table.

1.3.1 Core IGP Verification

The neighborship in the core router and their reachability are verified as follows:

```
regress@PE1> show ospf neighbor
Address Interface State ID Pri Dead
15.1.1.2 ae0.0 Full 6.6.6.6 128 37
15.1.2.2 ae1.0 Full 6.6.6.6 128 31
regress@PE1> show route protocol ospf table inet.0

inet.0: 60031 destinations, 60038 routes (60031 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

12.1.1.2/32 *[OSPF/10/10] 1d 17:40:52, metric 1
                    to 15.1.1.2 via ae0.0
                  > to 15.1.2.2 via ae1.0
12.1.1.3/32 *[OSPF/10/10] 1d 17:40:52, metric 2
                    to 15.1.1.2 via ae0.0
                  > to 15.1.2.2 via ae1.0
15.2.1.0/24 *[OSPF/10/10] 1d 17:40:52, metric 2
                    to 15.1.1.2 via ae0.0
                  > to 15.1.2.2 via ae1.0
15.2.2.0/24 *[OSPF/10/10] 1d 17:40:52, metric 2
                    to 15.1.1.2 via ae0.0
                  > to 15.1.2.2 via ae1.0
224.0.0.5/32 *[OSPF/10] 1d 17:41:08, metric 1
                    MultiRecv
224.0.0.6/32 *[OSPF/10] 1d 17:41:08, metric 1
                    MultiRecv

regress@PE1> show route protocol ospf table inet.3

inet.3: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

12.1.1.2/32 *[L-OSPF/10/5] 00:13:20, metric 1
                    to 15.1.1.2 via ae0.0
                  > to 15.1.2.2 via ae1.0
12.1.1.3/32 *[L-OSPF/10/5] 1d 17:40:55, metric 2
                    to 15.1.1.2 via ae0.0, Push 45103
                  > to 15.1.2.2 via ae1.0, Push 45103

regress@PE1> show mpls interface
Interface State Administrative groups (x: extended)
ae0.0 Up <none>
ae1.0 Up <none>

regress@PE1>
```

1.3.2 Core BGP Verification

```
regress@PE1> show bgp summary
Threading mode: BGP I/O
Default eBGP mode: advertise - accept, receive - accept
Groups: 2502 Peers: 2503 Down peers: 0
Table Tot Paths Act Paths Suppressed History Damp State Pending
inet.0
                60009 60002 0 0 0 0
inet6.0
                30002 30002 0 0 0 0
bgp.l3vpn.0
               580000 580000 0 0 0 0
```

```
bgp.l3vpn-inet6.0
                61700 61700 0 0 0 0
inet6.3
                   0 0 0 0 0 0
Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
12.1.1.3 65000 21313 20562 0 0 1d 17:40:57 Establ
  inet.0: 30002/30009/30009/0
  inet6.0: 15002/15002/15002/0
  bgp.l3vpn.0: 580000/580000/580000/0
  bgp.l3vpn-inet6.0: 61700/61700/61700/0
  METRO_VRF_1.inet.0: 251/251/251/0
  METRO_VRF_1.inet6.0: 101/101/101/0
  METRO_VRF_2.inet.0: 251/251/251/0
  METRO_VRF_2.inet6.0: 101/101/101/0
  METRO_VRF_3.inet.0: 251/251/251/0
  METRO_VRF_3.inet6.0: 101/101/101/0
...
  METRO_VRF_1997.inet.0: 251/251/251/0
  METRO_VRF_1998.inet.0: 251/251/251/0
  METRO_VRF_1999.inet.0: 251/251/251/0
  METRO_VRF_2000.inet.0: 251/251/251/0
  METRO_VRF_2001.inet.0: 51/51/51/0
  METRO_VRF_2002.inet.0: 51/51/51/0
  METRO_VRF_2003.inet.0: 51/51/51/0
...
  METRO_VRF_2998.inet.0: 51/51/51/0
  METRO_VRF_2999.inet.0: 51/51/51/0
  METRO_VRF_3000.inet.0: 51/51/51/0
  METRO_VRF_3001.inet.0: 52/52/52/0
  METRO_VRF_3002.inet.0: 52/52/52/0
  METRO_VRF_3003.inet.0: 52/52/52/0
...
  METRO_VRF_3498.inet.0: 52/52/52/0
  METRO_VRF_3499.inet.0: 52/52/52/0
  METRO_VRF_3500.inet.0: 52/52/52/0
  METRO_VRF_3501.inet.0: 2/2/2/0
  METRO_VRF_3502.inet.0: 2/2/2/0
  METRO_VRF_3503.inet.0: 2/2/2/0
...
  METRO_VRF_3998.inet.0: 2/2/2/0
  METRO_VRF_3999.inet.0: 2/2/2/0
  METRO_VRF_4000.inet.0: 2/2/2/0
  METRO_VRF_2001.inet6.0: 51/51/51/0
  METRO_VRF_2002.inet6.0: 51/51/51/0
  METRO_VRF_2003.inet6.0: 51/51/51/0
...
  METRO_VRF_2098.inet6.0: 51/51/51/0
  METRO_VRF_2099.inet6.0: 51/51/51/0
  METRO_VRF_2100.inet6.0: 51/51/51/0
  METRO_VRF_3001.inet6.0: 12/12/12/0
  METRO_VRF_3002.inet6.0: 12/12/12/0
  METRO_VRF_3003.inet6.0: 12/12/12/0
...
  METRO_VRF_3548.inet6.0: 2/2/2/0
  METRO_VRF_3549.inet6.0: 2/2/2/0
  METRO_VRF_3550.inet6.0: 2/2/2/0
16.0.1.2 45001 5003 5046 0 0 1d 17:39:49 Establ
  METRO_VRF_1.inet.0: 250/250/250/0
16.0.2.2 45002 5003 5046 0 0 1d 17:39:49 Establ
  METRO_VRF_2.inet.0: 250/250/250/0
16.0.3.2 45003 5003 5046 0 0 1d 17:39:54 Establ
  METRO_VRF_3.inet.0: 250/250/250/0
...
2002:16:0:1f3::2 45499 5003 5046 0 0 1d 17:39:32 Establ
  METRO_VRF_499.inet6.0: 100/100/100/0
2002:16:0:1f4::2 45500 5003 5046 0 0 1d 17:39:32 Establ
  METRO_VRF_500.inet6.0: 100/100/100/0
2002:16:100:1::2 65001 5036 5080 0 0 1d 17:39:59 Establ
  inet6.0: 15000/15000/15000/0

regress@PE1>
```

1.3.3 BGPv4 CE-PE Verification

The BGPv4 neighborship in one of the routing-instances METRO_VRF_1 is as follows:

```
regress@PE1> show bgp neighbor 16.0.1.2
Peer: 16.0.1.2+33773 AS 45001 Local: 16.0.1.1+179 AS 65000
  Group: METRO_VRF_EBGPv4_1 Routing-Instance: METRO_VRF_1
  Forwarding routing-instance: METRO_VRF_1
  Type: External State: Established Flags: <Sync>
  Last State: OpenConfirm Last Event: RecvKeepAlive
  Last Error: None
  Options: <AddressFamily PeerAS Refresh As Override>
  Options: <GracefulShutdownRcv>
  Address families configured: inet-unicast
  Holdtime: 90 Preference: 170
  Graceful Shutdown Receiver local-preference: 0
  Number of flaps: 0
  Receive eBGP Origin Validation community: Reject
  Peer ID: 192.0.0.1 Local ID: 16.0.1.1 Active Holdtime: 90
  Keepalive Interval: 30 Group index: 2 Peer index: 0 SNMP index: 3
  I/O Session Thread: bgpio-0 State: Enabled
  BFD: disabled, down
  Local Interface: et-0/0/0.1
  NLRI for restart configured on peer: inet-unicast
  NLRI advertised by peer: inet-unicast inet-multicast inet-vpn-unicast inet6-unicast inet6-
      multicast l2vpn inet6-vpn-unicast
  NLRI for this session: inet-unicast
  Peer supports Refresh capability (2)
  Stale routes from peer are kept for: 300
  Peer does not support Restarter functionality
  NLRI of all end-of-rib markers sent: inet-unicast
  Peer does not support Receiver functionality
  Peer does not support LLGR Restarter or Receiver functionality
  Peer does not support 4 byte AS extension
  Peer does not support Addpath
  Table METRO_VRF_1.inet.0 Bit: 60001
    RIB State: BGP restart is complete
    RIB State: VPN restart is complete
    Send state: in sync
    Active prefixes: 250
    Received prefixes: 250
    Accepted prefixes: 250
    Suppressed due to damping: 0
    Advertised prefixes: 251
  Last traffic (seconds): Received 1 Sent 18 Checked 154592
  Input messages: Total 5157 Updates 1 Refreshes 0 Octets 99082
  Output messages: Total 5201 Updates 2 Refreshes 0 Octets 99895
  Output Queue[5]: 0 (METRO_VRF_1.inet.0, inet-unicast)

regress@PE1>
```

The BGPv4 CE-PE routes in the routing-instance METRO_VRF_1 are as follows:

```
regress@PE1> show route table METRO_VRF_1.inet.0

METRO_VRF_1.inet.0: 503 destinations, 503 routes (503 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

16.0.1.0/24 *[Direct/0] 1d 18:03:51
                    > via et-0/0/0.1
16.0.1.1/32 *[Local/0] 1d 18:03:51
                      Local via et-0/0/0.1
17.0.1.0/24 *[BGP/170] 1d 17:53:50, localpref 100, from 12.1.1.3
                    AS path: I, validation-state: unverified
                  > to 15.1.1.2 via ae0.0, Push 16, Push 45103(top)
                    to 15.1.2.2 via ae1.0, Push 16, Push 45103(top)
20.0.1.0/24 *[BGP/170] 1d 17:52:42, localpref 100
                    AS path: 45001 I, validation-state: unverified
                  > to 16.0.1.2 via et-0/0/0.1
20.0.2.0/24 *[BGP/170] 1d 17:52:42, localpref 100
                    AS path: 45001 I, validation-state: unverified
                  > to 16.0.1.2 via et-0/0/0.1
...
20.0.249.0/24 *[BGP/170] 1d 17:52:42, localpref 100
                    AS path: 45001 I, validation-state: unverified
                  > to 16.0.1.2 via et-0/0/0.1
20.0.250.0/24 *[BGP/170] 1d 17:52:42, localpref 100
                    AS path: 45001 I, validation-state: unverified
                  > to 16.0.1.2 via et-0/0/0.1
30.0.1.0/24 *[BGP/170] 1d 17:52:39, localpref 100, from 12.1.1.3
                    AS path: 45001 I, validation-state: unverified
                    to 15.1.1.2 via ae0.0, Push 16, Push 45103(top)
                  > to 15.1.2.2 via ae1.0, Push 16, Push 45103(top)
30.0.2.0/24 *[BGP/170] 1d 17:52:39, localpref 100, from 12.1.1.3
```

```
                  AS path: 45001 I, validation-state: unverified
                   to 15.1.1.2 via ae0.0, Push 16, Push 45103(top)
                > to 15.1.2.2 via ae1.0, Push 16, Push 45103(top)
...
30.0.249.0/24 *[BGP/170] 1d 17:52:39, localpref 100, from 12.1.1.3
                  AS path: 45001 I, validation-state: unverified
                > to 15.1.1.2 via ae0.0, Push 16, Push 45103(top)
                   to 15.1.2.2 via ae1.0, Push 16, Push 45103(top)
30.0.250.0/24 *[BGP/170] 1d 17:52:39, localpref 100, from 12.1.1.3
                  AS path: 45001 I, validation-state: unverified
                > to 15.1.1.2 via ae0.0, Push 16, Push 45103(top)
                   to 15.1.2.2 via ae1.0, Push 16, Push 45103(top)

regress@PE1>
```

1.3.4 BGPv6 CE-PE Verification

The BGPv6 neighborship in one of the routing-instance METRO_VRF_1:

```
regress@PE1> show bgp neighbor 2002:16:0:1::2
Peer: 2002:16:0:1::2+32777 AS 45001 Local: 2002:16:0:1::1+179 AS 65000
  Group: METRO_VRF_EBGPv6_1 Routing-Instance: METRO_VRF_1
  Forwarding routing-instance: METRO_VRF_1
  Type: External State: Established Flags: <Sync>
  Last State: OpenConfirm Last Event: RecvKeepAlive
  Last Error: None
  Options: <AddressFamily PeerAS Refresh As Override>
  Options: <GracefulShutdownRcv>
  Address families configured: inet6-unicast
  Holdtime: 90 Preference: 170
  Graceful Shutdown Receiver local-preference: 0
  Number of flaps: 0
  Receive eBGP Origin Validation community: Reject
  Peer ID: 193.0.0.1 Local ID: 16.0.1.1 Active Holdtime: 90
  Keepalive Interval: 30 Group index: 3 Peer index: 0 SNMP index: 4
  I/O Session Thread: bgpio-0 State: Enabled
  BFD: disabled, down
  Local Interface: et-0/0/0.1
  NLRI for restart configured on peer: inet6-unicast
  NLRI advertised by peer: inet-unicast inet-multicast inet-vpn-unicast inet6-unicast inet6-
      multicast l2vpn inet6-vpn-unicast
  NLRI for this session: inet6-unicast
  Peer supports Refresh capability (2)
  Stale routes from peer are kept for: 300
  Peer does not support Restarter functionality
  NLRI of all end-of-rib markers sent: inet6-unicast
  Peer does not support Receiver functionality
  Peer does not support LLGR Restarter or Receiver functionality
  Peer does not support 4 byte AS extension
  Peer does not support Addpath
  NLRI that peer supports extended nexthop encoding for: inet-unicast inet-multicast inet-vpn-
      unicast
  Table METRO_VRF_1.inet6.0 Bit: 70001
    RIB State: BGP restart is complete
    RIB State: VPN restart is complete
    Send state: in sync
    Active prefixes: 100
    Received prefixes: 100
    Accepted prefixes: 100
    Suppressed due to damping: 0
    Advertised prefixes: 101
  Last traffic (seconds): Received 29 Sent 24 Checked 154980
  Input messages: Total 5169 Updates 1 Refreshes 0 Octets 99266
  Output messages: Total 5213 Updates 2 Refreshes 0 Octets 100103
  Output Queue[6]: 0 (METRO_VRF_1.inet6.0, inet6-unicast)

regress@PE1>
```

The BGPv6 CE-PE routes in the routing-instance METRO_VRF_1:

```
regress@PE1> show route table METRO_VRF_1.inet6.0

METRO_VRF_1.inet6.0: 205 destinations, 205 routes (205 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

```

```
2002:16:0:1::/64 *[Direct/0] 1d 18:07:23
                  > via et-0/0/0.1
2002:16:0:1::1/128 *[Local/0] 1d 18:07:23
                     Local via et-0/0/0.1
2002:17:0:1::/64 *[BGP/170] 1d 17:57:32, localpref 100, from 12.1.1.3
                   AS path: I, validation-state: unverified
                  > to 15.1.1.2 via ae0.0, Push 16, Push 2, Push 45103(top)
                    to 15.1.2.2 via ae1.0, Push 16, Push 2, Push 45103(top)
2002:20:0:1::/64 *[BGP/170] 1d 17:56:28, localpref 100
                   AS path: 45001 I, validation-state: unverified
                  > to 2002:16:0:1::2 via et-0/0/0.1
2002:20:0:2::/64 *[BGP/170] 1d 17:56:28, localpref 100
                   AS path: 45001 I, validation-state: unverified
                  > to 2002:16:0:1::2 via et-0/0/0.1
...
2002:20:0:63::/64 *[BGP/170] 1d 17:56:28, localpref 100
                   AS path: 45001 I, validation-state: unverified
                  > to 2002:16:0:1::2 via et-0/0/0.1
2002:20:0:64::/64 *[BGP/170] 1d 17:56:28, localpref 100
                   AS path: 45001 I, validation-state: unverified
                  > to 2002:16:0:1::2 via et-0/0/0.1
2002:30:0:1::/64 *[BGP/170] 1d 17:56:27, localpref 100, from 12.1.1.3
                   AS path: 45001 I, validation-state: unverified
                  > to 15.1.1.2 via ae0.0, Push 16, Push 2, Push 45103(top)
                    to 15.1.2.2 via ae1.0, Push 16, Push 2, Push 45103(top)
2002:30:0:2::/64 *[BGP/170] 1d 17:56:27, localpref 100, from 12.1.1.3
                   AS path: 45001 I, validation-state: unverified
                  > to 15.1.1.2 via ae0.0, Push 16, Push 2, Push 45103(top)
                    to 15.1.2.2 via ae1.0, Push 16, Push 2, Push 45103(top)
...
2002:30:0:63::/64 *[BGP/170] 1d 17:56:27, localpref 100, from 12.1.1.3
                   AS path: 45001 I, validation-state: unverified
                  > to 15.1.1.2 via ae0.0, Push 16, Push 2, Push 45103(top)
                    to 15.1.2.2 via ae1.0, Push 16, Push 2, Push 45103(top)
2002:30:0:64::/64 *[BGP/170] 1d 17:56:27, localpref 100, from 12.1.1.3
                   AS path: 45001 I, validation-state: unverified
                  > to 15.1.1.2 via ae0.0, Push 16, Push 2, Push 45103(top)
                    to 15.1.2.2 via ae1.0, Push 16, Push 2, Push 45103(top)
fe80::ae78:d100:12d:3504/128
                 *[Local/0] 1d 18:07:23
                    Local via et-0/0/0.1
ff02::2/128 *[INET6/0] 1d 18:10:03
                    MultiRecv

regress@PE1>
```

1.3.5 CE-PE OSPF Verification

OSPF Neighborship of the routing-instance is as shown below:

```
regress@PE1> show ospf neighbor instance METRO_VRF_2001
Address Interface State ID Pri Dead
16.10.1.2 et-0/0/2.1 Full 195.0.0.1 0 30
```

The OSPF routes received from the CE can be listed as:

```
regress@PE1> show route table METRO_VRF_2001.inet.0 protocol ospf

METRO_VRF_2001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

40.0.1.0/24 *[OSPF/10] 1d 16:50:57, metric 1
                > to 16.10.1.2 via et-0/0/2.1
40.0.2.0/24 *[OSPF/10] 1d 16:50:57, metric 1
                > to 16.10.1.2 via et-0/0/2.1
...
40.0.49.0/24 *[OSPF/10] 1d 16:50:57, metric 1
                > to 16.10.1.2 via et-0/0/2.1
40.0.50.0/24 *[OSPF/10] 1d 16:50:57, metric 1
                > to 16.10.1.2 via et-0/0/2.1
224.0.0.5/32 *[OSPF/10] 1d 18:13:21, metric 1
                  MultiRecv
224.0.0.6/32 *[OSPF/10] 1d 18:13:21, metric 1
                  MultiRecv

```

```
regress@PE1>
```

The OSPF received from the PE2 routes as BGP routes over the L3VPN BGP family can be listed as follows.

```
regress@PE1> show route table METRO_VRF_2001.inet.0 protocol bgp

METRO_VRF_2001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

17.10.1.0/24 *[BGP/170] 1d 18:02:09, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 1131, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 1131, Push 45103(top)
42.0.1.0/24 *[BGP/170] 1d 17:24:52, MED 1, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 1131, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 1131, Push 45103(top)
42.0.2.0/24 *[BGP/170] 1d 17:24:52, MED 1, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 1131, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 1131, Push 45103(top)
...
42.0.49.0/24 *[BGP/170] 1d 17:24:52, MED 1, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 1131, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 1131, Push 45103(top)
42.0.50.0/24 *[BGP/170] 1d 17:24:52, MED 1, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 1131, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 1131, Push 45103(top)

regress@PE1>
```

1.3.6 CE-PE OSPFv3 Verification

OSPFv3 Neighborship of the routing-instance is as shown below:

```
regress@PE1> show ospf3 neighbor instance METRO_VRF_2001
ID Interface State Pri Dead
209.0.0.1 et-0/0/2.1 Full 0 38
  Neighbor-address fe80::228:1ff:fe00:1

regress@PE1>
```

The OSPFv3 routes received from the CE can be listed as

```
regress@PE1> show route table METRO_VRF_2001.inet6.0 protocol ospf3

METRO_VRF_2001.inet6.0: 107 destinations, 107 routes (107 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

3000:7:1:1::/64 *[OSPF3/10] 1d 17:28:54, metric 1
              > to fe80::228:1ff:fe00:1 via et-0/0/2.1
3000:7:1:2::/64 *[OSPF3/10] 1d 17:28:54, metric 1
              > to fe80::228:1ff:fe00:1 via et-0/0/2.1
...
3000:7:1:31::/64 *[OSPF3/10] 1d 17:28:54, metric 1
              > to fe80::228:1ff:fe00:1 via et-0/0/2.1
3000:7:1:32::/64 *[OSPF3/10] 1d 17:28:54, metric 1
              > to fe80::228:1ff:fe00:1 via et-0/0/2.1
ff02::5/128 *[OSPF3/10] 1d 18:19:22, metric 1
                MultiRecv
ff02::6/128 *[OSPF3/10] 1d 18:19:22, metric 1
                MultiRecv

regress@PE1>
```

The OSPFv3 received from the PE2 routes as BGP routes over the 6VPE BGP family can be listed as follows.

```
regress@PE1> show route table METRO_VRF_2001.inet6.0 protocol bgp

METRO_VRF_2001.inet6.0: 107 destinations, 107 routes (107 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both
```

```
2002:17:10:1::/64 *[BGP/170] 1d 18:07:53, localpref 100, from 12.1.1.3
                    AS path: I, validation-state: unverified
                  > to 15.1.1.2 via ae0.0, Push 1131, Push 2, Push 45103(top)
                    to 15.1.2.2 via ae1.0, Push 1131, Push 2, Push 45103(top)
3000:6:1:1::/64 *[BGP/170] 1d 17:30:42, MED 1, localpref 100, from 12.1.1.3
                    AS path: I, validation-state: unverified
                  > to 15.1.1.2 via ae0.0, Push 1131, Push 2, Push 45103(top)
                    to 15.1.2.2 via ae1.0, Push 1131, Push 2, Push 45103(top)
...
3000:6:1:31::/64 *[BGP/170] 1d 17:30:42, MED 1, localpref 100, from 12.1.1.3
                    AS path: I, validation-state: unverified
                  > to 15.1.1.2 via ae0.0, Push 1131, Push 2, Push 45103(top)
                    to 15.1.2.2 via ae1.0, Push 1131, Push 2, Push 45103(top)
3000:6:1:32::/64 *[BGP/170] 1d 17:30:42, MED 1, localpref 100, from 12.1.1.3
                    AS path: I, validation-state: unverified
                  > to 15.1.1.2 via ae0.0, Push 1131, Push 2, Push 45103(top)
                    to 15.1.2.2 via ae1.0, Push 1131, Push 2, Push 45103(top)

regress@PE1>
```

1.3.7 CE-PE ISIS Verification

ISIS Adjacency of the routing-instance is as shown below:

```
regress@PE1> show isis adjacency instance METRO_VRF_3001
Interface System L State Hold (secs) SNPA
et-0/0/3.1 6401.0001.0000 2 Up 29

regress@PE1>
```

The ISIS routes received from the CE:

```
regress@PE1> show route table METRO_VRF_3001.inet.0 protocol isis

METRO_VRF_3001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

44.0.1.0/24 *[IS-IS/18] 1d 18:10:17, metric 10
                  > to 16.14.1.2 via et-0/0/3.1
44.0.2.0/24 *[IS-IS/18] 1d 18:10:17, metric 10
                  > to 16.14.1.2 via et-0/0/3.1
...
44.0.49.0/24 *[IS-IS/18] 1d 18:10:17, metric 10
                  > to 16.14.1.2 via et-0/0/3.1
44.0.50.0/24 *[IS-IS/18] 1d 18:10:17, metric 10
                  > to 16.14.1.2 via et-0/0/3.1

regress@PE1>
```

The ISIS routes received from the PE2 as BGP routes over the L3VPN BGP family can be listed as follows.

```
regress@PE1> show route table METRO_VRF_3001.inet.0 protocol bgp

METRO_VRF_3001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

12.4.0.1/32 *[BGP/170] 1d 18:16:45, localpref 100, from 12.1.1.3
                    AS path: I, validation-state: unverified
                    to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
                  > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
17.14.1.0/24 *[BGP/170] 1d 18:16:45, localpref 100, from 12.1.1.3
                    AS path: I, validation-state: unverified
                    to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
                  > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
45.0.1.0/24 *[BGP/170] 1d 18:14:58, MED 10, localpref 100, from 12.1.1.3
                    AS path: I, validation-state: unverified
                    to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
                  > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
45.0.2.0/24 *[BGP/170] 1d 18:14:58, MED 10, localpref 100, from 12.1.1.3
                    AS path: I, validation-state: unverified
                    to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
                  > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
...
```

```
45.0.49.0/24 *[BGP/170] 1d 18:14:58, MED 10, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
45.0.50.0/24 *[BGP/170] 1d 18:14:58, MED 10, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)

regress@PE1>
```

1.3.8 CE-PE ISISv6 Verification

ISISv6 Adjacency of the routing-instance is as shown below:

```
regress@PE1> show isis adjacency instance METRO_VRF_3001
Interface System L State Hold (secs) SNPA
et-0/0/3.1 6401.0001.0000 2 Up 29

regress@PE1>
```

The ISISv6 routes received from the CE can be listed as

```
regress@PE1> show route table METRO_VRF_3001.inet6.0 protocol isis

METRO_VRF_3001.inet6.0: 28 destinations, 28 routes (28 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

3000:4:1:1::/64 *[IS-IS/18] 1d 18:17:26, metric 10
              > to fe80::215:1ff:fe00:1 via et-0/0/3.1
3000:4:1:2::/64 *[IS-IS/18] 1d 18:17:26, metric 10
              > to fe80::215:1ff:fe00:1 via et-0/0/3.1
...
3000:4:1:9::/64 *[IS-IS/18] 1d 18:17:43, metric 10
              > to fe80::215:1ff:fe00:1 via et-0/0/3.1
3000:4:1:a::/64 *[IS-IS/18] 1d 18:17:43, metric 10
              > to fe80::215:1ff:fe00:1 via et-0/0/3.1

regress@PE1>
```

The ISISv6 routes received from the PE2 as BGP routes over the 6vpe bgp family can be listed as follows.

```
regress@PE1> show route table METRO_VRF_3001.inet6.0 protocol bgp

METRO_VRF_3001.inet6.0: 28 destinations, 28 routes (28 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2002:17:14:1::/64 *[BGP/170] 1d 18:20:01, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
              > to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
                to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
2005:12:4::1/128 *[BGP/170] 1d 18:20:01, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
              > to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
                to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
3001:1:1:1::/64 *[BGP/170] 1d 18:18:18, MED 10, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
              > to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
                to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
3001:1:1:2::/64 *[BGP/170] 1d 18:18:18, MED 10, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
              > to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
                to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
...
3001:1:1:9::/64 *[BGP/170] 1d 18:18:18, MED 10, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
3001:1:1:a::/64 *[BGP/170] 1d 18:18:18, MED 10, localpref 100, from 12.1.1.3
                AS path: I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)

regress@PE1>
```

1.3.9 CE-PE Static Verification

The Static routes configured for the CE can be listed as

```
regress@PE1> show route table METRO_VRF_3501.inet.0 protocol static

METRO_VRF_3501.inet.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

46.0.1.0/24 *[Static/5] 1d 18:33:39
                > to 16.16.1.2 via et-0/0/4.1

regress@PE1>
```

The Static routes received from the PE2 as BGP routes over the L3VPN BGP family can be listed as follows.

```
regress@PE1> show route table METRO_VRF_3501.inet.0 protocol bgp

METRO_VRF_3501.inet.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

17.16.1.0/24 *[BGP/170] 1d 18:24:25, localpref 100, from 12.1.1.3
                  AS path: I, validation-state: unverified
                > to 15.1.1.2 via ae0.0, Push 2797, Push 45103(top)
                  to 15.1.2.2 via ae1.0, Push 2797, Push 45103(top)
46.10.1.0/24 *[BGP/170] 1d 18:24:25, localpref 100, from 12.1.1.3
                  AS path: I, validation-state: unverified
                > to 15.1.1.2 via ae0.0, Push 2797, Push 45103(top)
                  to 15.1.2.2 via ae1.0, Push 2797, Push 45103(top)

regress@PE1>
```

1.3.10 CE-PE Staticv6 Verification

The Static-v6 routes received from the CE can be listed as

```
regress@PE1> show route table METRO_VRF_3501.inet6.0 protocol static

METRO_VRF_3501.inet6.0: 7 destinations, 7 routes (7 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2008:1:0:1::/64 *[Static/5] 1d 18:35:38
                > to 2002:16:16:1::2 via et-0/0/4.1

regress@PE1>
```

The Static-v6 routes received from the PE2 as BGP routes over the L3VPN BGP family can be listed as follows.

```
regress@PE1> show route table METRO_VRF_3501.inet.0 protocol bgp

METRO_VRF_3501.inet.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

17.16.1.0/24 *[BGP/170] 1d 18:26:14, localpref 100, from 12.1.1.3
                  AS path: I, validation-state: unverified
                > to 15.1.1.2 via ae0.0, Push 2797, Push 45103(top)
                  to 15.1.2.2 via ae1.0, Push 2797, Push 45103(top)
46.10.1.0/24 *[BGP/170] 1d 18:26:14, localpref 100, from 12.1.1.3
                  AS path: I, validation-state: unverified
                > to 15.1.1.2 via ae0.0, Push 2797, Push 45103(top)
                  to 15.1.2.2 via ae1.0, Push 2797, Push 45103(top)

regress@PE1>
```

1.3.11 6PE Verification

The neighborship of the BGP Global IPv6 Peer is listed below

```
regress@PE1> show bgp neighbor 2002:16:100:1::2
Peer: 2002:16:100:1::2+32777 AS 65001 Local: 2002:16:100:1::1+179 AS 65000
  Group: EBGP Routing-Instance: master
  Forwarding routing-instance: master
  Type: External State: Established Flags: <Sync>
  Last State: OpenConfirm Last Event: RecvKeepAlive
  Last Error: None
  Options: <AddressFamily PeerAS Refresh As Override>
  Options: <GracefulShutdownRcv>
  Address families configured: inet6-unicast
  Holdtime: 90 Preference: 170
  Graceful Shutdown Receiver local-preference: 0
  Number of flaps: 0
  Receive eBGP Origin Validation community: Reject
  Peer ID: 201.0.0.1 Local ID: 12.1.1.1 Active Holdtime: 90
  Keepalive Interval: 30 Group index: 1 Peer index: 0 SNMP index: 2
  I/O Session Thread: bgpio-0 State: Enabled
  BFD: disabled, down
  Local Interface: et-0/0/1.4000
  NLRI for restart configured on peer: inet6-unicast
  NLRI advertised by peer: inet-unicast inet-multicast inet-vpn-unicast inet6-unicast inet6-
      multicast l2vpn inet6-vpn-unicast
  NLRI for this session: inet6-unicast
  Peer supports Refresh capability (2)
  Stale routes from peer are kept for: 300
  Peer does not support Restarter functionality
  NLRI of all end-of-rib markers sent: inet6-unicast
  Peer does not support Receiver functionality
  Peer does not support LLGR Restarter or Receiver functionality
  Peer does not support 4 byte AS extension
  Peer does not support Addpath
  NLRI that peer supports extended nexthop encoding for: inet-unicast inet-multicast inet-vpn-
      unicast
  Table inet6.0 Bit: 30001
    RIB State: BGP restart is complete
    Send state: in sync
    Active prefixes: 15000
    Received prefixes: 15000
    Accepted prefixes: 15000
    Suppressed due to damping: 0
    Advertised prefixes: 15002
  Last traffic (seconds): Received 21 Sent 16 Checked 147443
  Input messages: Total 4951 Updates 34 Refreshes 0 Octets 231303
  Output messages: Total 4994 Updates 36 Refreshes 0 Octets 232009
  Output Queue[2]: 0 (inet6.0, inet6-unicast)

regress@PE1>
```

The 6PE routes received from the Global Peer and Remote Peer PE2 is as shown below:

```
regress@PE1> show route table inet6.0

inet6.0: 30008 destinations, 30008 routes (30008 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

2002:12:1:1::1/128 *[Direct/0] 1d 17:10:57
                    > via lo0.0
2002:12:1:1::3/128 *[BGP/170] 1d 16:59:24, localpref 100, from 12.1.1.3
                      AS path: I, validation-state: unverified
                    > to 15.1.1.2 via ae0.0, Push 2, Push 45103(top)
                      to 15.1.2.2 via ae1.0, Push 2, Push 45103(top)
2002:16:100:1::/64 *[Direct/0] 1d 17:09:13
                    > via et-0/0/1.4000
2002:16:100:1::1/128
                   *[Local/0] 1d 17:09:13
                      Local via et-0/0/1.4000
2002:17:100:1::/64 *[BGP/170] 1d 16:59:24, localpref 100, from 12.1.1.3
                      AS path: I, validation-state: unverified
                    > to 15.1.1.2 via ae0.0, Push 2, Push 45103(top)
                      to 15.1.2.2 via ae1.0, Push 2, Push 45103(top)
2002:50:0:1::/64 *[BGP/170] 1d 16:58:26, localpref 100
                    AS path: 65001 I, validation-state: unverified
                  > to 2002:16:100:1::2 via et-0/0/1.4000
2002:50:0:2::/64 *[BGP/170] 1d 16:58:26, localpref 100
                    AS path: 65001 I, validation-state: unverified
                  > to 2002:16:100:1::2 via et-0/0/1.4000
2002:50:0:3::/64 *[BGP/170] 1d 16:58:26, localpref 100
                    AS path: 65001 I, validation-state: unverified
                  > to 2002:16:100:1::2 via et-0/0/1.4000
```

```
...
2002:55:0:3a96::/64*[BGP/170] 1d 16:59:11, localpref 100, from 12.1.1.3
                AS path: 65001 I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 4018, Push 2, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 4018, Push 2, Push 45103(top)
2002:55:0:3a97::/64*[BGP/170] 1d 16:59:11, localpref 100, from 12.1.1.3
                AS path: 65001 I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 4018, Push 2, Push 45103(top)
              > to 15.1.2.2 via ae1.0, Push 4018, Push 2, Push 45103(top)
2002:55:0:3a98::/64*[BGP/170] 1d 16:59:11, localpref 100, from 12.1.1.3
                AS path: 65001 I, validation-state: unverified
              > to 15.1.1.2 via ae0.0, Push 4018, Push 2, Push 45103(top)
                 to 15.1.2.2 via ae1.0, Push 4018, Push 2, Push 45103(top)
fe80::ae78:d10f:a02d:350c/128
             *[Local/0] 1d 17:10:02
                Local via et-0/0/1.4000
fe80::ae78:d1f0:2d:3503/128
             *[Direct/0] 1d 17:11:44
              > via lo0.0
ff02::2/128 *[INET6/0] 1d 17:12:42
                MultiRecv

regress@PE1>
```

1.3.12 Global IPv4 Verification

The neighborship of the BGP Global IPv4 Peer is listed below:

```
regress@PE1> show bgp neighbor 16.100.1.2
Peer: 16.100.1.2+32779 AS 65001 Local: 16.100.1.1+179 AS 65000
  Group: EBGP Routing-Instance: master
  Forwarding routing-instance: master
  Type: External State: Established Flags: <Sync>
  Last State: OpenConfirm Last Event: RecvKeepAlive
  Last Error: None
  Options: <AddressFamily PeerAS Refresh As Override>
  Options: <GracefulShutdownRcv>
  Address families configured: inet-unicast
  Holdtime: 90 Preference: 170
  Graceful Shutdown Receiver local-preference: 0
  Number of flaps: 0
  Receive eBGP Origin Validation community: Reject
  Peer ID: 200.0.0.1 Local ID: 12.1.1.1 Active Holdtime: 90
  Keepalive Interval: 30 Group index: 1 Peer index: 1 SNMP index: 1
  I/O Session Thread: bgpio-0 State: Enabled
  BFD: disabled, down
  Local Interface: et-0/0/1.4000
  NLRI for restart configured on peer: inet-unicast
  NLRI advertised by peer: inet-unicast inet-multicast inet-vpn-unicast inet6-unicast inet6-
      multicast l2vpn inet6-vpn-unicast
  NLRI for this session: inet-unicast
  Peer supports Refresh capability (2)
  Stale routes from peer are kept for: 300
  Peer does not support Restarter functionality
  NLRI of all end-of-rib markers sent: inet-unicast
  Peer does not support Receiver functionality
  Peer does not support LLGR Restarter or Receiver functionality
  Peer does not support 4 byte AS extension
  Peer does not support Addpath
  Table inet.0 Bit: 20001
    RIB State: BGP restart is complete
    Send state: in sync
    Active prefixes: 30000
    Received prefixes: 30000
    Accepted prefixes: 30000
    Suppressed due to damping: 0
    Advertised prefixes: 30002
  Last traffic (seconds): Received 17 Sent 4 Checked 147619
  Input messages: Total 4953 Updates 30 Refreshes 0 Octets 215047
  Output messages: Total 4998 Updates 34 Refreshes 0 Octets 215788
  Output Queue[1]: 0 (inet.0, inet-unicast)

regress@PE1>
```

The IPv4 routes received from the Global Peer and Remote Peer PE2 is as shown below:

```
regress@PE1> show route table inet.0 protocol bgp

inet.0: 60057 destinations, 60090 routes (60057 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

...

50.0.1.0/24 *[BGP/170] 1d 19:15:32, localpref 100
                AS path: 65001 I, validation-state: unverified
              > to 16.100.1.2 via et-0/0/1.4000
50.0.2.0/24 *[BGP/170] 1d 19:15:32, localpref 100
                AS path: 65001 I, validation-state: unverified
              > to 16.100.1.2 via et-0/0/1.4000
...

55.117.47.0/24 *[BGP/170] 1d 19:16:33, localpref 100, from 12.1.1.3
                AS path: 65001 I, validation-state: unverified
                 to 15.1.1.2 via ae0.0, Push 45103
              > to 15.1.2.2 via ae1.0, Push 45103
55.117.48.0/24 *[BGP/170] 1d 19:16:33, localpref 100, from 12.1.1.3
                AS path: 65001 I, validation-state: unverified
              > to 15.1.1.2 via ae0.0, Push 45103
                to 15.1.2.2 via ae1.0, Push 45103
...

regress@PE1>
```

1.4 Verifying Feature At Scale

The following output captures BGP, OSPF & OSPFv3 peering scale:

```
regress@PE1> show bgp summary |match Estab | count
Count: 2503 lines

regress@PE1> show ospf neighbor instance all |match Full | count
Count: 1002 lines

regress@PE1> show ospf3 neighbor instance all |match Full | count
Count: 100 lines
```

```
regress@PE1> show route summary
Autonomous system number: 65000
Router ID: 12.1.1.1

Highwater Mark (All time / Time averaged watermark)
   RIB unique destination routes: 2032662 at 2022-10-26 21:14:00 / 2032660
   RIB routes : 2032671 at 2022-10-26 21:14:00 / 2032669
   FIB routes : 1377491 at 2022-10-25 04:57:29 / 1377491
   VRF type routing instances : 4000 at 2022-10-25 03:35:04

inet.0: 60031 destinations, 60038 routes (60031 active, 0 holddown, 0 hidden)
           Direct: 5 routes, 5 active
            Local: 4 routes, 4 active
             OSPF: 6 routes, 6 active
              BGP: 60009 routes, 60002 active
           Static: 14 routes, 14 active

inet.3: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
           L-OSPF: 2 routes, 2 active

METRO_VRF_1.inet.0: 503 destinations, 503 routes (503 active, 0 holddown, 0 hidden)
           Direct: 1 routes, 1 active
            Local: 1 routes, 1 active
              BGP: 501 routes, 501 active

METRO_VRF_2.inet.0: 503 destinations, 503 routes (503 active, 0 holddown, 0 hidden)
           Direct: 1 routes, 1 active
            Local: 1 routes, 1 active
              BGP: 501 routes, 501 active
...
METRO_VRF_2001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
           Direct: 1 routes, 1 active
            Local: 1 routes, 1 active
             OSPF: 52 routes, 52 active
              BGP: 51 routes, 51 active
```

```
...
METRO_VRF_3001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
           Direct: 2 routes, 2 active
            Local: 1 routes, 1 active
              BGP: 52 routes, 52 active
            IS-IS: 50 routes, 50 active
...
METRO_VRF_3501.inet.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)
           Direct: 1 routes, 1 active
            Local: 1 routes, 1 active
              BGP: 2 routes, 2 active
           Static: 1 routes, 1 active
...
mpls.0: 4014 destinations, 4014 routes (4014 active, 0 holddown, 0 hidden)
             MPLS: 6 routes, 6 active
              VPN: 4001 routes, 4001 active
           L-OSPF: 7 routes, 7 active

bgp.l3vpn.0: 580000 destinations, 580000 routes (580000 active, 0 holddown, 0 hidden)
              BGP: 580000 routes, 580000 active

inet6.0: 30008 destinations, 30008 routes (30008 active, 0 holddown, 0 hidden)
           Direct: 3 routes, 3 active
            Local: 2 routes, 2 active
              BGP: 30002 routes, 30002 active
           INET6: 1 routes, 1 active

inet6.3: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
           L-OSPF: 2 routes, 2 active

METRO_VRF_1.inet6.0: 205 destinations, 205 routes (205 active, 0 holddown, 0 hidden)
           Direct: 1 routes, 1 active
            Local: 2 routes, 2 active
              BGP: 201 routes, 201 active
           INET6: 1 routes, 1 active
...
METRO_VRF_2001.inet6.0: 107 destinations, 107 routes (107 active, 0 holddown, 0 hidden)
           Direct: 1 routes, 1 active
            Local: 2 routes, 2 active
           OSPF3: 52 routes, 52 active
              BGP: 51 routes, 51 active
           INET6: 1 routes, 1 active
...
METRO_VRF_3001.inet6.0: 28 destinations, 28 routes (28 active, 0 holddown, 0 hidden)
           Direct: 3 routes, 3 active
            Local: 2 routes, 2 active
              BGP: 12 routes, 12 active
           IS-IS: 10 routes, 10 active
           INET6: 1 routes, 1 active
...
METRO_VRF_3501.inet6.0: 7 destinations, 7 routes (7 active, 0 holddown, 0 hidden)
           Direct: 1 routes, 1 active
            Local: 2 routes, 2 active
              BGP: 2 routes, 2 active
           Static: 1 routes, 1 active
           INET6: 1 routes, 1 active
...
bgp.l3vpn-inet6.0: 61700 destinations, 61700 routes (61700 active, 0 holddown, 0 hidden)
              BGP: 61700 routes, 61700 active

regress@PE1>
```

1.4.1 Traffic Verification at Router at PE1

Traffic from the CE flows towars the core over the two LAG bundles ae0 and ae1 and traffic load-balanced across the bundles and the member interfaces of the bundle.

```
PE1 Seconds: 6 Time: 20:32:01

Interface Link Input packets (pps) Output packets (pps)

 ae0 Up 5586190244536 (38943485) 5575357824202 (38871357)
 ae1 Up 5555772552639 (38763433) 5566579800978 (38841649)
 esi Up 0 (0) 0 (0)
 et-0/0/0 Up 2228214479834 (15511380) 2228214835244 (15513576)
 et-0/0/1 Up 2229235576443 (15520844) 2229235532415 (15521822)
```

```
et-0/0/2 Up 2228332663250 (15517930) 2227956502538 (15518860)
et-0/0/3 Up 2228318956042 (15518420) 2228336007745 (15519712)
et-0/0/4 Up 2228317159060 (15523076) 2228317066058 (15524683)
et-0/0/5 Down 0 (0) 0 (0)
et-0/0/6 Up 1410137465761 (9827780) 1390248083194 (9689916)
et-0/0/7 Up 1405078195401 (9794287) 1397493349083 (9742324)
et-0/0/8 Up 1369119218901 (9545572) 1392702964826 (9710874)
et-0/0/9 Up 1401855364473 (9775846) 1394913427099 (9728243)
et-0/0/10 Up 1392259351671 (9710934) 1391134693656 (9703807)
et-0/0/11 Up 1389924844295 (9696679) 1390350321922 (9700313)
et-0/0/12 Up 1381150822629 (9637431) 1395562427879 (9738708)
et-0/0/13 Up 1392437534044 (9718389) 1389532357521 (9698821)
```

1.5 Generating Scale Configuration

The Python Script is having three components, a Jinja File capturing the various configuration templates, a Params File capturing the various user variables and the script uses both these input files and generates the required configurations and commit that in the router.

Note: The configs are generated as per the CE-bound interface schema.

1.5.1 Jinja Configuration Template

The template contains configurations for both interfaces which are part of the routing instances and routing instance configurations.

```
# Routing Instnace Configuration
# File Name: l3vpn_jinja.j2
# Version: 1.0

{%- if ce_pe_is_bgpv4 %}
groups {{group_name}} {
   routing-instances {
      {%- for vrf_id, ifl_name, peeras, neighborip4 in l3vpn_variables %}
      METRO_VRF_{{vrf_id}} {
         instance-type vrf;
         interface {{ifl_name}};
         route-distinguisher {{router_id}}:{{vrf_id}};
         vrf-target target:{{local_as_no}}:{{vrf_id}};
         vrf-table-label;
         protocols {
            bgp {
               group METRO_VRF_EBGPv4_{{vrf_id}} {
                  type external;
                  family inet {
                     unicast {
                     }
                  }
                  peer-as {{peeras}};
                  as-override;
                  neighbor {{neighborip4}};
               }
            }
         }
      }
      {%- endfor %}
   }
}
{%- endif %}
{%- if ce_pe_is_bgpv6 %}
groups {{group_name}} {
   routing-instances {
      {%- for vrf_id, ifl_name, peeras, neighborip6 in vpnv6_variables %}
      METRO_VRF_{{vrf_id}} {
         instance-type vrf;
         interface {{ifl_name}};
         route-distinguisher {{router_id}}:{{vrf_id}};
         vrf-target target:{{local_as_no}}:{{vrf_id}};
         vrf-table-label;
         protocols {
```

```
            bgp {
               group METRO_VRF_EBGPv6_{{vrf_id}} {
                  type external;
                  family inet6 {
                     unicast {
                     }
                  }
                  peer-as {{peeras}};
                  as-override;
                  neighbor {{neighborip6}};
               }
            }
         }
      }
      {%- endfor %}
   }
}
{%- endif %}
{%- if ce_pe_is_ospf %}
groups {{group_name}} {
   routing-instances {
      {%- for vrf_id, ifl_name in ospf_variables %}
      METRO_VRF_{{vrf_id}} {
         instance-type vrf;
         interface {{ifl_name}};
         route-distinguisher {{router_id}}:{{vrf_id}};
         vrf-target target:{{local_as_no}}:{{vrf_id}};
         vrf-table-label;
         protocols {
            ospf {
               area 0.0.0.0 {
                  interface {{ifl_name}} {
                     interface-type p2p;
                  }
               }
            }
         }
      }
      {%- endfor %}
   }
}
{%- endif %}
{%- if ce_pe_is_ospfv6 %}
groups {{group_name}} {
   routing-instances {
      {%- for vrf_id, ifl_name in ospf_variables %}
      METRO_VRF_{{vrf_id}} {
         instance-type vrf;
         interface {{ifl_name}};
         route-distinguisher {{router_id}}:{{vrf_id}};
         vrf-target target:{{local_as_no}}:{{vrf_id}};
         vrf-table-label;
         protocols {
            ospf3 {
               area 0.0.0.0 {
                  interface {{ifl_name}} {
                     interface-type p2p;
                  }
               }
            }
         }
      }
      {%- endfor %}
   }
}
{%- endif %}
{%- if ce_pe_is_isis %}
groups {{group_name}} {
   interfaces {
      lo0 {
         {%- for unit_id, ip_addr, iso_addr in lo0_variables %}
         unit {{unit_id}} {
            family inet {
               address {{ip_addr}};
            }
            family iso {
               address {{iso_addr}};
            }
         }
         {%- endfor %}
      }
```

```
    }
    routing-instances {
       {%- for vrf_id, ifl_name, lo0_ifl in isis_variables %}
       METRO_VRF_{{vrf_id}} {
          instance-type vrf;
          interface {{ifl_name}};
          interface {{lo0_ifl}};
          route-distinguisher {{router_id}}:{{vrf_id}};
          vrf-target target:{{local_as_no}}:{{vrf_id}};
          vrf-table-label;
          protocols {
             isis {
                interface {{ifl_name}} {
                   level 1 disable;
                   point-to-point;
                }
                interface {{lo0_ifl}} {
                   passive;
                }
             }
          }
       }
       {%- endfor %}
    }
}
{%- endif %}
{%- if ce_pe_is_isisv6 %}
groups {{group_name}} {
   interfaces {
      lo0 {
         {%- for unit_id, ipv6_addr, iso_addr in lo0v6_variables %}
         unit {{unit_id}} {
            family inet6 {
               address {{ipv6_addr}};
            }
            family iso {
               address {{iso_addr}};
            }
         }
         {%- endfor %}
      }
   }
   routing-instances {
      {%- for vrf_id, ifl_name, lo0_ifl in isisv6_variables %}
      METRO_VRF_{{vrf_id}} {
         instance-type vrf;
         interface {{ifl_name}};
         interface {{lo0_ifl}};
         route-distinguisher {{router_id}}:{{vrf_id}};
         vrf-target target:{{local_as_no}}:{{vrf_id}};
         vrf-table-label;
         protocols {
            isis {
               interface {{ifl_name}} {
                  level 1 disable;
                  point-to-point;
               }
               interface {{lo0_ifl}} {
                  passive;
               }
            }
         }
      }
      {%- endfor %}
   }
}
{%- endif %}
{%- if ce_pe_is_static %}
groups {{group_name}} {
   routing-instances {
      {%- for vrf_id, ifl_name, static_route, next_hop_value in static_variables %}
      METRO_VRF_{{vrf_id}} {
         instance-type vrf;
         interface {{ifl_name}};
         route-distinguisher {{router_id}}:{{vrf_id}};
         vrf-target target:{{local_as_no}}:{{vrf_id}};
         vrf-table-label;
         routing-options {
            static {
               route {{static_route}} next-hop {{next_hop_value}};
            }
```

```
            }
        }
        {%- endfor %}
    }
}
{%- endif %}
{%- if ce_pe_is_staticv6 %}
groups {{group_name}} {
   routing-instances {
      {%- for vrf_id, ifl_name, static_routev6, next_hop_valuev6 in staticv6_variables %}
      METRO_VRF_{{vrf_id}} {
         instance-type vrf;
         interface {{ifl_name}};
         route-distinguisher {{router_id}}:{{vrf_id}};
         vrf-target target:{{local_as_no}}:{{vrf_id}};
         vrf-table-label;
         routing-options {
            rib METRO_VRF_{{vrf_id}}.inet6.0 {
               static {
                  route {{static_routev6}} next-hop {{next_hop_valuev6}};
               }
            }
         }
      }
      {%- endfor %}
   }
}
{%- endif %}
```

1.5.2 Configuration Parameters

‘‘‘



DC – AI – Routing – Switching – Security – Wireless – Automation

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